Data Mining Project M2

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***Abstract*—This document discusses the current plans and progress regarding an analysis of a dataset of 24 million Google reviews for this semester's Data Mining project. This project aims to answer three interrelated discovery questions, two of which are related to discovering groups of reviewers and businesses respectively. The last asks which users and restaurants exhibit anomalous behavior.**

I. Introduction

In today's world we have a lot of data. For example, Google Maps provides us with the location of almost every business, street, and house in the U.S. It also contains a catalog of reviews for each business. In the pursuit of equipping myself with a greater understanding of what Data Mining is, how it works, and how it is useful, I am using a dataset related to these reviews to find useful and interesting patterns.

II. Dataset Description

The dataset I originally used did not have metadata included, nor could I find the metadata online. As a result, I decided to choose a dataset that was closely related in topic. There was a different dataset for each U.S. State, the District of Columbia, and other places within U.S. jurisdiction. For each dataset there was a JSON file for the data and a JSON file for the metadata. I decided to only use the Georgia dataset because it alone was 7.5 Gigabytes and it included 24 million reviews for 166,381 businesses.

The metadata contains:

- Business Name
- Address
- Google Maps ID
- Description
- Latitude
- Longitude
- Category
- Average Rating
- Number of Reviews
- Price
- Hours
- Miscellaneous Information
- State (Whether it's open)
- Relative Results

The data includes:

- User ID
- User name
- Microsecond the review was posted
- Rating
- Review text
- Pictures included with the review
- Responses to the review
- Google Maps ID of the business reviewed

I will be using all the attributes within the metadata except for the ones listed below:

- Business Name: Not useful for most circumstances, but this could change.
- Average Rating: We can calculate average rating with our regular data
- Number of Reviews: We already can calculate this in the data
- Price: This is usually null.
- Miscellaneous Information: Inconsistent subset of information
- State (Whether it's open): Redundant/useless information
- Relative Results: Might eventually use to compare to the results of the second discovery question.

Similarly, I will be using all the attributes within the data except the user name, which is redundant.

{"name": "Porter Pharmacy", "address": "Porter Pharmacy, 129 N Second St, Cochran, GA 31014", "gmap_id": "0x88f16e41928ff687:0x883dad4fd048e8f8", "description": null, "latitude": 32.3883, "longitude": -83.3571, "category": ["Pharmacy"], "avg_rating": 4.9, "num_of_reviews": 16, "price": null, "hours": [["Friday", "8AM\u20136PM"], ["Saturday", "8AM\u201312PM"], ["Sunday", "Closed"], ["Monday", "8AM\u20136PM"], ["Tuesday", "8AM\u20136PM"], ["Wednesday", "8AM\u201312PM"], ["Thursday", "8AM\u20136PM"]], "MISC": {"Service options": ["In-store shopping", "Same-day delivery"], "Health & safety": ["Mask required", "Staff required to disinfect surfaces between visits"], "Accessibility": ["Wheelchair accessible entrance"], "Planning": ["Quick visit"]}, "state": "Open \u22c5 Closes 6PM", "relative_results": ["0x88f16e41929435cf:0x5b2532a2885e9ef6", "0x88f16c32716531c1:0x5f19bdaa5044e4fa", "0x88f16e6e3f4a21df:0xcf495da9bb4d89ea"], "url": "https://www.google.com/maps/place//data=!4m2!3m1!1s0x88f16e41928ff687:0x883dad4fd048e8f8?authuser=-1&hl=en&gl=us"}
{"name": "Communication Dynamics Pediatric Therapy", "address": "Communication Dynamics Pediatric Therapy, 1385 Oglethorpe St, Macon, GA 31201", "gmap_id": "0x88f3fe84b3eb91a9:0xc632096c7ee64b71", "description": null, "latitude": 32.834362, "longitude": -83.64522, "category": ["Speech pathologist"], "avg_rating": 4.8, "num_of_reviews": 18, "price": null, "hours": [["Thursday", "8:30AM\u20136PM"], ["Friday", "Closed"], ["Saturday", "Closed"], ["Sunday", "Closed"], ["Monday", "8:30AM\u20136PM"], ["Tuesday", "8:30AM\u20136PM"], ["Wednesday", "8:30AM\u20136PM"]], "MISC": {"From the business": ["Identifies as Black-owned", "Identifies as women-led"], "Health & safety": ["Appointment required", "Mask required", "Temperature check required", "Staff wear masks", "Staff get temperature checks"], "Accessibility": ["Wheelchair accessible entrance", "Wheelchair accessible restroom"], "Amenities": ["Restroom"], "Planning": ["Appointments recommended"]}, "state": "Open \u22c5 Closes 6PM", "relative_results": ["0x88f3f8a4f7618ee9:0x44e17d62eefc6bc9", "0x88f3f8a4f7618ee9:0xc1031b7ffbbffed2", "0x88f4079dc1a62807:0x509accf85f196646", "0x88f3f8a4f7618ee9:0xd9ac489f4db29ea4", "0x88f3e6d407a5ae4f:0xa1ce83e302b7e296"], "url": "https://www.google.com/maps/place//data=!4m2!3m1!1s0x88f3fe84b3eb91a9:0xc632096c7ee64b71?authuser=-1&hl=en&gl=us"}

Fig. 1. Sample of the Metadata

```
{"user_id": "104227684932200174741", "name": "Brent Powell", "time": 1599604812269,
 "rating": 5, "text": "Great hometown folk. Rob takes good care of all of us. We choose
 hometown over bigger any day of the week! Keep up the good work!", "pics": null, "resp":
 null, "gmap_id": "0x88f16e41928ff687:0x883dad4fd048e8f8"}
{"user_id": "101391439865171313417", "name": "Lisa Daws", "time": 1597063645403, "rating"
  5, "text": "So helpful with my dad's needs when he was terminally ill. They also do
 Medicare plan consulting which is helpful in the area.", "pics": null, "resp": null,
 "gmap_id": "0x88f16e41928ff687:0x883dad4fd048e8f8"}
{"user_id": "117805059305497697012", "name": "Tanya S. Manning", "time": 1572145871222,
 "rating": 5, "text": "It is a comfort to know your pharmacist always has you and your
 family's best interests at heart. No matter how busy he is, Rob never fails to stop and
 answer any questions we have!", "pics": null, "resp": {"time": 1572297435382, "text": "H
  Tanya, we're forever grateful for your kind words and appreciate the time you took to
 leave behind a review! We strive to treat our patients like family and happy to help
 anytime."}, "gmap_id": "0x88f16e41928ff687:0x883dad4fd048e8f8"}
{"user_id": "115776338830397150217", "name": "James Hagler", "time": 1557920569537,
 "rating": 5, "text": "Friendly staff and awesomely helpful.  Forgetful me left my
 medicine when I went out of town and Rob was able to find a local pharmacy where I was
 that would get me pills for a low cost for the days I was gone.  Local delivery as well
 that is always quick.", "pics": null, "resp": null, "gmap_id":
 "0x88f16e41928ff687:0x883dad4fd048e8f8"}
{"user_id": "117505875431546709002", "name": "JAN BROWN", "time": 1504822535645, "rating"
  5, "text": "We are new residents  in Cochran and the new  owners were so very helpful
 and kind! They remember who you are and honestly care about your health... not like the
 *corporate* places here in town, these people want to help and advise you... I am more
 than a number to them, I am a human with medical needs and concerns...  plus they have
 the cutest nick-nacks, just like the old time pharmacies!", "pics": null, "resp": null,
 "gmap_id": "0x88f16e41928ff687:0x883dad4fd048e8f8"}
```


Fig. 2. Sample of the Data

There are a variety of data quality issues within the dataset. For instance, there are 986 cases where at least the name, address, and Google Maps ID match another instance (effectively being a duplicate). Similarly, there are 322,360 duplicated reviews in the data, which could slightly skew the results.

In the metadata there are improperly entered latitudes and longitudes. There is at least one case where the longitude of a business is 180°. This longitude is in the middle of the Pacific Ocean, which is not located near Georgia. In addition, there are other cases where the latitude and/or longitude listed does not intersect with Georgia.

There are attributes which are so inconsistent that I would be unable to make use of them at all. For instance, the price in the metadata is usually null, making it completely useless.

III. Discovery Questions

The discovery questions include:

- What natural segments of reviewers emerge based on rating behavior, review frequency, temporal activity patters, and picture-posting behavior?
- What groups of businesses share similar review distribution patterns and engagement characteristics?
- Which users or restaurants exhibit anomalous behavior in rating patterns, timing, or review structure?

The first discovery question groups reviewers together by their behavioral patterns. I intend on using clustering (DBSCAN) to answer this question, using each user's:

- total number of reviews
- total number of pictures
- total number of words
- coefficient of variation regarding time between each review
- coefficient of variation regarding ratings

to form a scatter plot to categorize users.

The second discovery question groups similar businesses by review characteristics. I will repeat mostly the same process as with the first discovery question, but using different variables. I might I will use the category of business. However, I do know that I will use the coefficient of variance regarding these:

- Number of pictures
- Number of words
- Time between each review
- Replies to ratings
- Ratings

This last discovery question combines the first two ideas to find anomalies within the data. In addition to these clustering methods with scatterplots, I shall use histograms on many of the attributes (and derived attributes) within the reviews to detect anomalies.

IV. Preprocessing

My first step was to search for issues within the dataset. Since I noticed that there were duplicate reviews in the dataset, I decided to check for duplicates in the metadata and data first. Since I did not have the memory to store every part of the data, I decided to instead scan for repeats of a handful of values that together, shouldn't repeat.

For the metadata I chose the business name, address, and Google Maps ID. I chose these three because no two businesses would share both a name and an address without being a duplicate. I also included the Google Maps ID just to be sure.

To detect duplicates in the data I decided to use the user ID, time of the review's posting, and Google Maps ID. I do not believe it to be possible that the same user would post multiple reviews to the same place because:

1) Google doesn't allow users to make reviews at the same place twice.
2) Even if this were the case, users wouldn't do this twice within a millisecond.
3) Someone programming a bot would be incentivized to use multiple accounts, not the same account in order to be convincing.

I was able to get a complete count of duplicates within both files with two different Python Scripts. Following this, I made a script to remove said duplicates along with labels I wouldn't use. In addition, I replaced the text and pictures field with the count of each respectively. My technique reduced the size of the metadata file by 74.39%, and the size of the data file by 56.5%.

I also made a script that took the cleaned review file and reduced it further to only hold the time, rating, picture count, and word count for the purpose of doing exploratory data analysis without causing my computer to run out of memory.

Later, I made another script to group the cleaned review file to hold statistics related to each user, such as the total number of pictures, reviews, and words they have written. This particular script was crucial for doing clustering on User Data.

V. Key EDA Insights

Here is a collection of tables relating to modified versions of various datasets:

Statistic	Latitude	Longitude	Num_of_Reviews
Count	165395.000000	165395.000000	165395.000000
Mean	33.432707	-83.858679	145.357109
Std	0.950851	2.404695	369.566285
Min	25.126484	-88.100026	1.000000
25%	33.041988	-84.472016	8.000000
50%	33.773949	-84.218039	35.000000
75%	34.000048	-83.658343	118.000000
Max	42.639763	180.000000	9998.000000

TABLE I
SUMMARY STATISTICS FOR SELECTED METADATA VARIABLES.

I removed the variable representing the average rating because this table wouldn't fit properly in the document otherwise. If you wish to see this information or other columns I will exclude, go look at my Jupyter Notebook file. Above

Statistic	Time	Pics	Word_Count
Count	23707799.00	23707799.00	23707799.00
Mean	1551798490370.16	0.06	17.34
Std	44324114719.32	0.73	38.19
Min	946598400000.00	0.00	0.00
25%	1523875184762.50	0.00	0.00
50%	1555365798320.00	0.00	4.00
75%	1582721919769.00	0.00	18.00
Max	1631158941334.00	50.00	2109.00

TABLE II
ULTRA-CLEANED DATA FOR FOUR VARIABLES

is a sample of review-Georgia_CandR.json. I made this JSON file so that I could run the describe method on a handful of columns in the data file without my computer running out of memory. Thanks to it, I was able to find anomalies in the number of pictures sent. I removed the rating attribute from the table. Below this there is a sample of the data.


```
{"time":1599604812269,"rating":5,"pics":
{"time":1597063645403,"rating":5,"pics":
{"time":1572145871222,"rating":5,"pics":
{"time":1557920569537,"rating":5,"pics":
```


Statistic	Total_Pictures	Max_Pics	Total_Reviews
Count	820735.00	820735.00	820735.00
Mean	0.79	0.29	10.92
Std	11.47	1.78	26.29
Min	0.00	0.00	1.00
25%	0.00	0.00	1.00 0
50%	0.00	0.00	2.00
75%	0.00	0.00	7.00
Max	2175.00	50.00	1520.00

TABLE III
GROUPED BY USER TABLE 1

This is the first of three tables from rgur_data.json. Its name stands for something approximating Reviews Grouped by User Reviews, but I forgot its original name. The Rev_With_Pics has been removed from this table. This table showed another interesting anomaly, that one user made 1,520 reviews, which was significantly higher than both the mean and median.

Statistic	Max_Words	Rev_With_Words	Avg_Pictures
Count	820735.00	820735.00	820735.00
Mean	57.17	6.37	0.04
Std	77.75	17.40	0.34
Min	0.00	0.00	0.00
25%	11.00	1.00	0.00
50%	34.00	2.00	0.00
75%	73.00	4.00	0.00
Max	1604.00	1506.00	49.00

TABLE IV
GROUPED BY USER TABLE 2

This table has Total_Words removed. Max words shows that most reviews are fairly short, but one particular review extends to 1,604 words. Further investigation needs to be done.

Statistic	Pics_Percent	Cov_Time	Cov_Rating
Count	820735.00	820735.00	820735.00
Mean	1.50	0.73	0.13
Std	8.55	1.08	0.20
Min	0.00	0.00	0.00
25%	0.00	0.00	0.00
50%	0.00	0.00	0.00
75%	0.00	1.18	0.23
Max	100.00	20.81	0.89

TABLE V
GROUPED BY USER TABLE 3

This last table shows the number of reviews containing pictures, the coefficient of variation for the users' time between reviews, and the coefficient of variation for the users' review scores. No column has been removed from this table.

VI. Mining Results

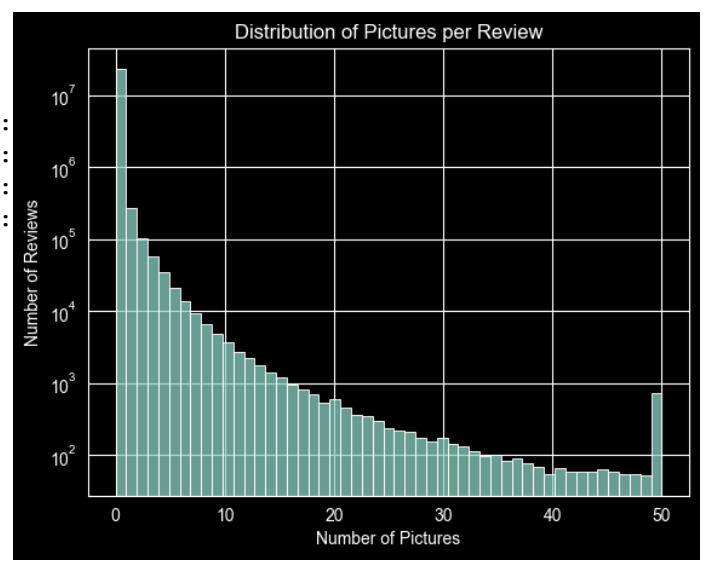

Fig. 3. Histogram for the number of pictures included in reviews.

Note: I had to make this histogram logarithmic in order for it to properly display information. As can be clearly observed, the overwhelming majority of reviews do not include any pictures, and there are more reviews with fewer pictures than

reviews with more pictures. Interestingly, there are approximately 10 times more reviews with 50 pictures than reviews with 49 pictures. I think the reason for this is that because the limit is 50 pictures, reviewers who would have posted even more pictures are forced into this number.

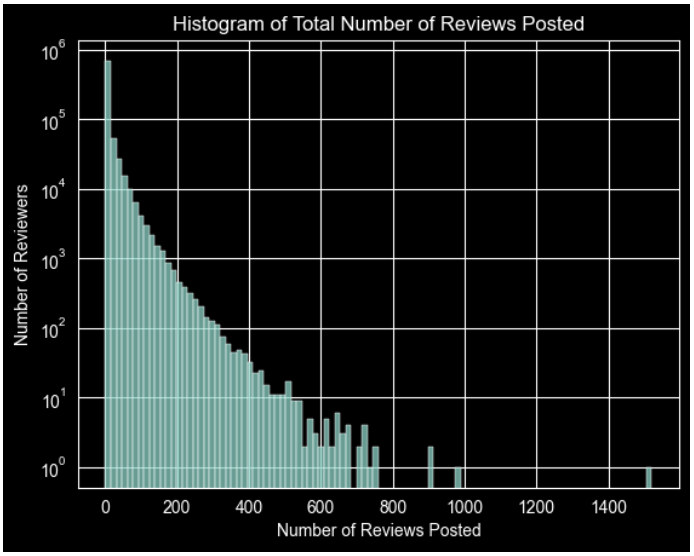

Fig. 4. Frequency distribution of individual user review counts.

This histogram is also logarithmically scaled. We can see that the one user that posted 1,520 reviews is a significant outlier, having posted more than 50% more reviews than even the second-highest review count.

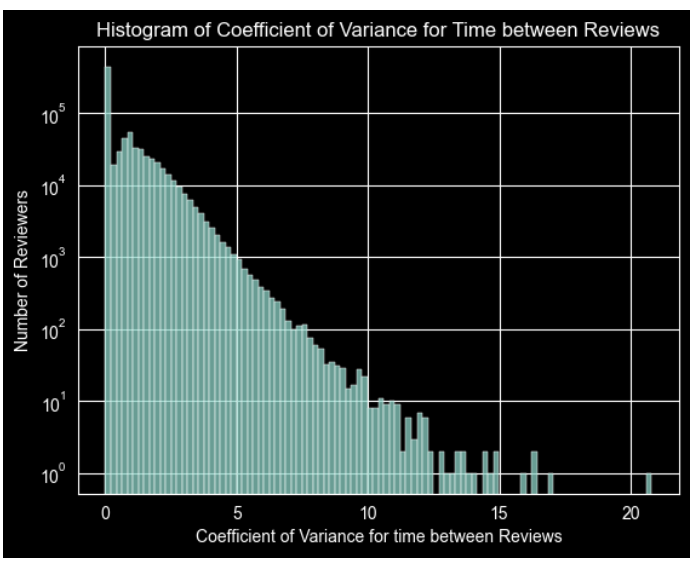

Fig. 5. Frequency distribution of coefficient of variance for time between reviews.

Note: Reviewers with only one review are automatically assigned a value of 0. I used the coefficient of variance, given by this formula:

$$CV = \frac{\sigma}{\mu}$$

This allows me to capture which reviewers wrote reviews more often, potentially allowing me to detect bots. I have done little else with this information.

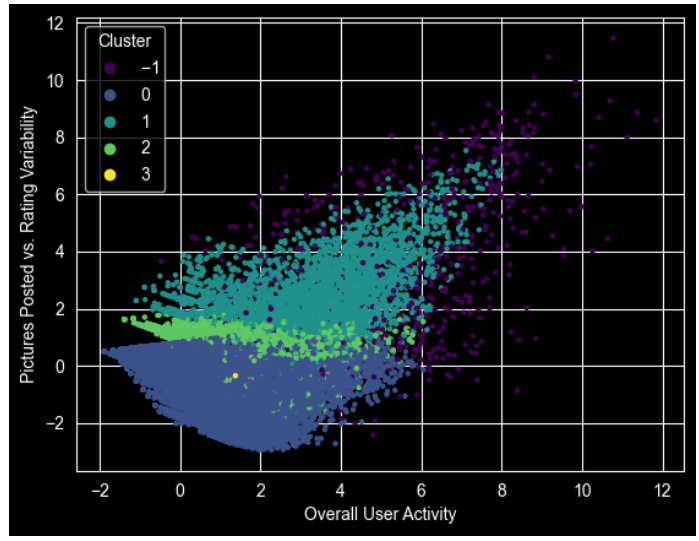

Fig. 6. Dimension-reduced scatterplot of DBSCAN algorithm run on 100,000 users.

More recently, I ran the DBSCAN algorithm on the first 100,000 users listed in rgur_data.json. The five variables I chose as dimensions were:

- Logarithmically-scaled total review count
- Logarithmically-scaled total picture count
- Logarithmically-scaled total word count
- Coefficient of Variance for the gap in time between reviews
- Coefficient of Variance of Ratings

DBSCAN is useful for both identifying outliers and classifying users. The only issue is that I have to decide on epsilon and minimum number of points. I decided upon a minimum number of points of 10 as it was double the number of dimensions. In addition, I used ScikitLearn's NearestNeighbors to compute the epsilon.

This created a scatterplot that poorly displayed its information. Take for example that the yellow cluster-set is barely visible because they are behind the other clusters (with one visible exception).

	log_reviews	log_pics	log_words	cov_time	cov_rating
PC1	0.5656	0.3184	0.4177	0.5249	0.3588
PC2	0.0788	0.7129	-0.2066	0.0972	-0.6584

TABLE VI
PRINCIPAL COMPONENT ANALYSIS LOADINGS

When I reduced the dimensions (because a 5D scatterplot can't be viewed on this plane of existence) I used Principle Component Analysis to automatically generate axis for the scatter plot. This applied weights to the five dimensions to create a visual that places each cluster in a different region. Unfortunately, it did not do a very ideal job. The cluster represented with yellow is extremely small in number, and are

completely surrounded by parts of the blue and green clusters. Here is the total count for each cluster:

TABLE VII
INDIVIDUAL CLUSTER COUNTS

Cluster ID	Count
0	92,874
1	3,864
2	2,620
-1	626
3	16

VII. CHANGES FROM M1 AND PLANS FOR M3

I have rewritten the majority of this project going from M1 to M2 for two key reasons:

1) My dataset lacked any metadata, leaving me unable to answer key discovery questions.
2) My questions were tendency-based, making them poor discovery questions

I had no solid plan going into this, so nothing of value was lost upon these revelations. The only thing that remains from my previous M1 was the idea of using a Google Maps dataset to find meaningful information relating to reviews.

My plans for M3 are also not very solid. This mostly has to do with the fact that I have been rushing this overdue project for the past four days. Since I have most work towards answering all three discovery questions almost complete, I will refine whatever I have and accomplish whatever new thing M3 asks of me to complete.

REFERENCES

[1] Google Dataset Citation 1 UCTopic: Unsupervised Contrastive Learning for Phrase Representations and Topic Mining Jiacheng Li, Jingbo Shang, Julian McAuley Annual Meeting of the Association for Computational Linguistics (ACL), 2022

[2] Google Dataset Citation 2 Personalized Showcases: Generating Multi-Modal Explanations for Recommendations An Yan, Zhankui He, Jiacheng Li, Tianyang Zhang, Julian Mcauley arXiv:2207.00422, 2022